Giant Beetle Larvae Are Quietly Eating Blueberry Roots

From: ScienceDaily Top Environment Mined: 2026-11-08 5 min read

North Carolina's blueberry growers are facing a new threat from a destructive beetle whose **larvae** feed underground, damaging plant roots and potentially killing blueberry bushes. Scientists have now identified the insect responsible, an important first step toward protecting crops from the pest.

A research team at North Carolina State University has confirmed that the culprit is Prionus imbricornus, a species of **longhorn beetle** distinguished by its long **antennae**. These wood-boring insects are becoming a growing concern for blueberry producers in the state.

Female beetles generally lay their eggs in soil near the roots of hardwood trees. After hatching, the **larvae** feed on the roots, sometimes destroying enough tissue to kill the tree. The **larvae** can reach an impressive five inches in length, while the adult beetles do not feed at all.

North Carolina is the first state to document P. imbricornus feeding on blueberry bushes. But signs of a similar problem have existed for years. Reports of unidentified Prionus **larvae** damaging blueberry roots date back to 2010, but researchers had struggled to determine which species was responsible.

Part of the challenge was the insects' underground habitat. Because the **larvae** remain near plant roots, they are difficult to locate and study. Making matters more complicated, **larvae** from different **longhorn beetle** species often look remarkably similar, making visual identification unreliable.

According to Kenneth Geisert, a graduate student at NC State and lead author of the study, this uncertainty has made it difficult to develop pest control strategies. "Before now, researchers often just assumed the species of Prionus on their commodities based on adult identification," Geisert said. "If that guess was incorrect, it could mean using a treatment strategy that did not line up with the problem and incorrectly associating species and their hosts. For instance, while P. imbricornus attacks roots, another type of **longhorn beetle** might attack the trunk of a tree, or dead branches. Without knowing which species of beetle you're dealing with and their ecology, incorrect management can cause adverse effects on **non-target insects**."

To identify the insects, the research team did fieldwork at six blueberry farms in North Carolina's Pender, Sampson, Bladen, and New Hanover counties. The scientists installed black panel traps containing sex **pheromones**, chemical signals that attract adult beetles. After collecting the insects, they used a DNA identification technique known as **genetic barcoding** to determine the identity of **larvae** found in the fields.

This method examines short, standardized sections of DNA and compares them with reference sequences from known species. By matching genetic material from the unidentified **larvae** with DNA from adult Prionus beetles, the researchers could determine which species was present. The analysis produced a 98 to 99 percent match with P. imbricornus, confirming the identity of the beetle behind the root damage. Identifying the pest is an important breakthrough, and it is also the first step toward finding the right way to fight

it.

Word Treasure

larvae -- The young of a beetle, worm-like in shape, before it grows into an adult.
longhorn beetle -- A beetle known for its very long antennae.
antennae -- The long feelers on an insect's head that it uses to sense the world around it.
pheromones -- Chemical signals that attract other members of the same species.
genetic barcoding -- A way of telling species apart by comparing short matching pieces of their DNA.
non-target insects -- Insects that a pest treatment is not meant to harm.

Background you need

Blueberry bushes are a crop, and like other crops they can be attacked by insect pests that eat their roots, stems, or fruit.

Longhorn beetles are a large family of insects named for their extra-long antennae, and many of them lay their eggs near trees so their larvae can feed on wood or roots underground.

Because the larvae live out of sight and look alike across species, scientists often need DNA tests to tell one kind from another before they can choose the right way to control it.

Knowing exactly which pest is present matters, because different beetles attack different parts of a plant and need different treatments.

Break it down

LEAD: North Carolina blueberry growers face a new pest: beetle larvae eating roots underground, and scientists have now named the culprit.

KEY EVENT: NC State researchers confirm the pest is the longhorn beetle Prionus imbricornus.

+- WHO: A research team at North Carolina State University, led by graduate student Kenneth Geisert.

+- WHAT: Larvae up to five inches long feed on blueberry roots and can kill bushes; adult beetles do not feed at all.

+- WHEN: Mystery root damage reported since 2010; North Carolina is the first state to document this species on blueberries.

+- HOW: Black panel traps with sex pheromones caught adult beetles, and genetic barcoding matched larvae DNA at 98 to 99 percent.

+- WHERE: Six blueberry farms in Pender, Sampson, Bladen, and New Hanover counties, North Carolina.

EVIDENCE: The DNA match of 98 to 99 percent with P. imbricornus, plus traps that attracted adult beetles.

OPEN QUESTION: How to fight the pest now that it is identified -- the right treatment still has to be found.

Why it matters

Knowing the exact pest lets farmers pick a treatment that works and avoids harming helpful insects, protecting a food crop and the farms that grow it.

Test what you remember

1. Which beetle species did the scientists identify as the culprit?

- (A) Prionus imbricornus
- (B) A beetle that only eats dead leaves
- (C) A honeybee
- (D) A ladybug

2. How long can the larvae grow?

- (A) Five inches
- (B) Two feet
- (C) Ten inches
- (D) One inch

3. How did the research team confirm which species the larvae belonged to?

- (A) By asking the farmers to guess
- (B) By looking at the color of the eggs
- (C) By genetic barcoding
- (D) By watching the adult beetles fly

4. In which North Carolina counties did the team do fieldwork?

- (A) Pender, Sampson, Bladen, and New Hanover counties
- (B) Six counties in South Carolina
- (C) Mecklenburg and Guilford counties
- (D) Wake, Durham, and Orange counties

5. Why was it so hard for researchers to tell which species was attacking the blueberries?

- (A) The larvae live underground and look alike across species
- (B) The larvae change color every day
- (C) The farms would not let scientists visit
- (D) The beetles only come out during snowstorms

6. What did Kenneth Geisert warn could happen if researchers guessed the wrong species?

- (A) The farms would have to close for a year
- (B) The beetles would learn to fly farther
- (C) A treatment could target the wrong pest and harm other insects
- (D) The blueberries would grow twice as big

Answer key (try the quiz first!): 1: A 2: A 3: C 4: A 5: A 6: C

Questions to think about

1. Kenneth Geisert, lead author of the study: Guessing the species from adult beetles can lead to the wrong treatment, which may fail against the real pest and harm insects that were never the target.

MY THOUGHTS (at least 20 words):